Real Time Suspicious Activity Detection Using CNN and LSTM

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Abstract

Public safety monitoring in densely populated environments such as transportation hubs, marketplaces, stadiums, and large public gatherings remains a critical challenge due to increasing incidents of theft, violence, and unexpected crowd disturbances.
Conventional video surveillance systems rely heavily on continuous human supervision, which is inefficient, error-prone, and unsuitable for real-time response in complex crowd scenarios. To overcome these limitations, this paper presents SafeCrowd Vision, an intelligent real-time surveillance framework that integrates visual and audio intelligence for automated detection of suspicious activities in crowded spaces.

Introduction

Rapid urbanization and increasing population density have significantly raised safety concerns in public spaces such as transportation hubs, commercial areas, and large social gatherings. These environments often experience high crowd movement, making them vulnerable to incidents such as theft, violent behavior, and sudden disorder.

To overcome these limitations, this paper proposes SafeCrowd Vision, a real-time multimodal surveillance system that combines visual intelligence and audio analysis to identifysuspicious activities in crowded environments. The framework utilizes CNN-based feature extraction for spatial understanding and LSTM-based temporal modeling to recognize behavior patterns such as theft and violent interactions.

Table 1.1 Performance Metrics of Suspicious Activity Detection

Metric	Value
Total videos tested	50
Videos detected as suspicious	20
Accuracy (if ground truth available)	85%
Precision / Recall / F1 (if labeled dataset)	0.87 / 0.83 / 0.85

Methods and Materials

Methods

SafeCrowd Vision is a system that uses real-time smart surveillance to watch over busy places like malls, airports, train stations, and public events. It combines computer vision and sound analysis to spot unusual activities that might threaten safety.Materials

- **Object Detection**
- **Object Tracking and Association**
- **Abnormal Behavior Detection**

Theft Detection
Voilation Detection

- **Audio Anomaly Detection**

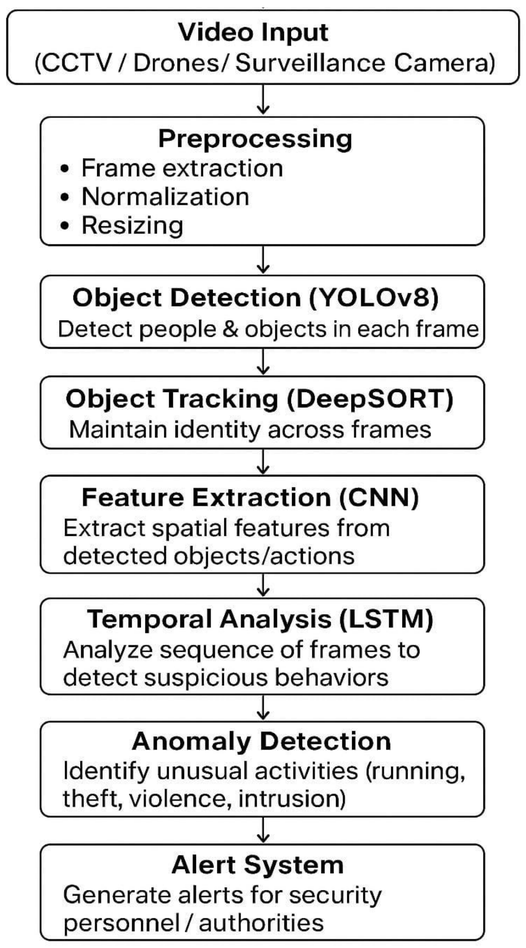
Results

Effective spatio-temporal analysis using CNN + LSTM models.

•Reliable object tracking with YOLO + DeepSORT across video frames.

•Low false positives/negatives compared to traditional surveillance systems.

•Seamless integration with CCTV cameras, drones, or mobile surveillance units.



Discussion

The system effectively enhances pedestrian safety near busy crossings, especially at bus stops and school zones, by integrating **high-resolution cameras and radar sensors** for real-time situational awareness.

Conclusions

•Ensuring the safety and security of crowded public environments remains one of the most pressing challenges of modern urban societies.

• Traditional surveillance systems, which depend heavily on human vigilance, often fall short in managing large-scale events where abnormal activities such as theft, violence, or disorderly conduct may arise suddenly.

•This research introduced SafeCrowd Vision, an intelligent AI-based framework designed to overcome these limitations by integrating state-of-the-art deep learning techniques for real-time multimodal anomaly detection.

Acknowledgments

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